AgroOrbit | EdgeOrbit AI

Satellite-to-Soil GeoAI for Clean Agricultural Pumping & Solar Irrigation Efficiency

AFRICA ENERGY INNOVATION CHALLENGE (AEIC 2026)
TOP 36 FINALIST • ALGERIA

01. The Energy-Water Problem in African Agriculture

Across rural Africa, agricultural water pumping accounts for over **35% of off-grid diesel and mini-grid electrical consumption**. Due to zero real-time soil data, farmers irrigate blind:

- **30%–40% Pumping Energy Wasted:** Farmers pump far more water than crops can absorb, draining diesel fuel budgets and depleting local aquifers.
- **Prohibitive Hardware Barrier:** Physical IoT capacitive soil probes cost over **€2,500 ($2,700) each**—completely unaffordable for fragmented smallholder farms.
- **Oversized Solar PV Systems:** Solar irrigation pumps are over-dimensioned by 40% to compensate for over-irrigation, bloating rural electrification CapEx.

02. The Technical Design: Zero-Hardware Satellite AI

AgroOrbit eliminates in-situ physical sensors by delivering automated, daily soil moisture intelligence derived directly from European Space Agency Copernicus satellites:

1. Copernicus Ingestion	2. Cloud-Penetrating SAR	3. Quantized AI Model	4. Actionable Alerts
Sentinel-1 Radar (SAR) + Sentinel-2 Optical	C-Band microwaves penetrate overcast & rain	INT8 CNN-Transformer on solar edge gateway	Precise m^3/ha irrigation advice on SMS/WhatsApp

- **All-Weather Reliability:** Radar penetrates clouds 365 days/year.
- **Sub-Field Precision:** 10m spatial resolution computed in **<45 ms**.
- **Ultra-Low Power Edge:** Runs on a <2W solar-sustained gateway.

-30%	$0	8x ROI
PUMPING FUEL & ENERGY WASTE	HARDWARE PROBE CAPEX REQUIRED	ANNUAL COST SAVINGS FOR FARMERS

03. Economic Feasibility & Quantified Benefits

- **Immediate Fuel Savings:** Average commercial plot of 50 ha saves over **3,800 Liters of diesel/year** (~$3,500 direct operating savings).
- **Solar Pumping Optimization:** Sizing solar arrays to exact agronomic crop demand reduces required PV panel & battery battery capacity by **25%–30%**, accelerating off-grid farm solarization.
- **Clean Energy ROI:** Payback period for the farmer is under **45 days** based purely on saved pumping electricity and diesel fuel.

04. Business Model & African Target Market

- **Per-Hectare SaaS:** Affordable subscription of **$5.00–$12.00 / hectare / year** for farmer cooperatives and agribusinesses.
- **B2G & Mini-Grid Licensing:** Enterprise dashboard licenses ($15k–$40k/yr) for Rural Electrification Agencies & River Basin Water Authorities.
- **Market Opportunity:** $1.8B agricultural water-energy market across 45 million irrigated hectares in North and Sub-Saharan Africa.
- **Traction:** 500 hectares in active validation; 5,000 ha pipeline.

05. Prototype Development & Field Testing Timeline (2026–2028)

Q3–Q4 2026 (Now)
Automated Sentinel-1C radar pipeline; 500 ha Algerian pilot (Mitidja & Chélif); INNOspace Masters & AEIC Top 36.

Q1–Q2 2027
Low-power solar edge gateway deployment; 3 farmer cooperative contracts (5,000 ha); Tunisian Medjerda Basin trial.

Q3–Q4 2027
Integration with solar irrigation pump controllers (smart relay switching); Morocco pilot (Souss-Massa); 25,000 ha.

2028 Scale-up
Pan-African expansion (Egypt, Kenya, Nigeria solar mini-grids); 100,000+ ha; $1M ARR.

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